

Auditing Counterfactual Data Augmentation in Reinforcement Learning

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Abstract. We audit a ground-up reimplementation of CTRL, which augments offline reinforcement-learning data by inferring and reusing transition-specific exogenous noise. The audit separates three questions that an end-to-end score conflates: whether the environment is learnable, whether exact noise reuse helps, and whether a learned structural causal model can approximate the corresponding counterfactuals. A reported-architecture D3QN attempt does not recover the CartPole performance plotted by Lu et al. A disclosed stabilized learner does recover the same numerical range, but is not a faithful reproduction. We then run a frozen, paired 30-seed mechanism study using exact simulator transitions. After matching synthetic-pool size, optimizer updates, and within-transition noise sharing, oracle reuse of factual noise scores 62.78 return points below an independently drawn shared-noise control on clean evaluation (95% paired bootstrap interval $[-111.31, -14.56]$). Under our process-noise stress test, the difference is -2.05 and meets a study-defined ± 5 practical-equivalence rule. Finally, a positive-weight BiCoGAN candidate fails held-out transition, termination, and latent-calibration gates in all five development seeds. Thus, instance-specific noise reuse is not sufficient to outperform the matched fresh-shared control in this setting, and the present implementation does not validate learned counterfactual augmentation.

Keywords: causal reinforcement learning · counterfactual augmentation · offline reinforcement learning · reproducibility

1 Introduction

Counterfactual augmentation promises to improve data efficiency without further environment interaction. In a structural causal model (SCM), dynamics are represented as

$$S_{t+1} = f(S_t, A_t, U_{t+1}), \quad (1)$$

where U_{t+1} contains exogenous factors. For an observed transition, a counterfactual first abducts its realized u_{t+1} and then changes the action while holding that realization fixed. This differs from sampling new noise, which produces another transition-distribution draw rather than an individual counterfactual [12,3].

CTRL applies this idea to offline reinforcement learning (RL): a Bidirectional Conditional Generative Adversarial Network (BiCoGAN) estimates the forward

mechanism and inverse noise mapping, after which a Dueling Double Deep Q-Network (D3QN) is trained on real and generated transitions [11,9]. Lu et al. report strong CartPole sample efficiency, but provide no official implementation and leave consequential details unspecified, including failure handling, generated data mixing, convergence criteria, and whether state noise is process or sensor noise. Our original manuscript treated exploratory, incompatible experiments as evidence for a broad conditional benefit. This revision withdraws those claims and asks a narrower, auditable question: *what does transition-specific noise reuse add when the simulator mechanism is known exactly?*

Our contributions are:

- a direct, protocol-qualified comparison with Lu et al., plus learner, environment, and protocol diagnostics that expose a failed reproduction;
- two separate paired 30-seed simulator studies, including a follow-up that matches noise sharing and isolates factual coupling from synthetic-pool diversity; and
- a held-out model-quality gate showing why the attempted learned BiCoGAN cannot support a downstream learned-counterfactual claim.

2 Background and Related Work

Lu et al. prove counterfactual identifiability when $f(s, a, \cdot)$ is strictly monotone and give Q-learning convergence results under their stated assumptions [11]. Standard neural networks do not provide this property automatically. Positive-weight and constrained monotonic networks are possible implementation directions [15], while recent work clarifies that counterfactual identification depends on recoverability of the exogenous structure [4].

The literature after CTRL also makes clear that augmentation quality and scope must be controlled. CoDA and MoCoDA exploit locally factored dynamics [13,14]; ACAMDA uses guided adversarial counterfactual estimation [16]; and CAIAC limits augmentation using causal action influence [2]. Causal state and transfer representations address related generalization questions [8,7]; Deng et al. provide a broader causal-RL survey [5]. Our study does not compete with these methods. It is a diagnostic of the simpler CTRL mechanism and the evidence required before a learned augmentation result is reported.

3 Audit Design

3.1 CartPole-SD protocol

We reconstruct the Simple Dataset (SD) setting described by Lu et al.: 250 random-action trials, at most 20 consecutive transitions per trial, 11 actions, and Gaussian state and action noise with standard deviation 0.05. A commanded action $a \in \{0, \dots, 10\}$ maps to $a/10$; after action noise ϵ_a , the applied force is $[2(a/10 + \epsilon_a) - 1]10$. State noise is added after one physics step. Our main

protocol feeds the noisy next state into subsequent physics, making it explicit *process noise*. Termination is computed from the pre-state-noise next state, and collection stops on failure. The resulting factual datasets in the frozen follow-up contain 3,365–3,705 transitions rather than exactly 5,000.

Clean evaluation uses the same 11-action simulator with both noise terms disabled. **Process-noise evaluation** uses both standard deviations at 0.05. Both have a 500-step ceiling. We do not call the latter the original “CTRL/noisy” protocol because Lu et al. do not specify evaluation noise. Each policy is evaluated on the same 100 episode seeds. Evaluation episodes are nested measurements; the training/data seed is the experimental unit.

Table 1 separates reported facts from choices that cannot be recovered from the original text. These gaps prevent a claim of exact implementation equivalence even when nominal sample count and architecture are matched.

Table 1. Reproduction target and disclosed protocol differences. NR means not reported by Lu et al.

Feature	Lu et al.	This audit
Collection	250 random-action trials; Same nominal design; failures 20 steps; 11 actions; 0.05 stop collection in the primary state/action noise	protocol
Factual transitions	Figure indexed by sample count; 5,000 shown	3,365–3,705 in the frozen follow-up; 5,000 tested by continuation diagnostic
State-noise semantics	Process versus sensor interpretation NR	Process noise primary; observation-noise sensitivity analysis
Evaluation	Ten trials; noise and seed policy NR	100 fixed episode seeds; clean and process-noise evaluations reported separately
Learner budget	Train to convergence	10,000 updates for architecture check; 5,000 for stabilized mechanism study
Generated-data mixing	NR	Fixed 50:50 real/generated minibatches, batch size 256
Downstream learner	Four 512-unit D3QN layers	Same architecture first; then disclosed 2x256 D3QN+CQL stabilization

3.2 Evidence tiers and learners

Development runs are separated from frozen multi-seed studies. First, we test the reported four-layer, 512-unit, batch-normalized D3QN architecture [17,6]. It fails a real-data learner gate. For the mechanism study only, we select a stabilized learner using real-data development runs: two 256-unit hidden layers, no batch

normalization, Adam with learning rate 10^{-4} , Polyak coefficient 0.005, 5,000 updates, and a Conservative Q-Learning (CQL) coefficient of 0.05 [10]. This deviation is central: it cannot rescue a faithful reproduction claim.

3.3 Noise-reuse controls

For each factual transition and each of its ten alternative actions, the exact simulator constructs:

- **Fresh-independent synthetic**: draw new action and state noise separately for each alternative action;
- **Fresh-shared synthetic**: draw one independent noise pair per factual transition and reuse it over all ten alternative actions; and
- **Oracle CF**: reuse the factual transition’s action and state noise over all ten alternative actions.

Fresh-shared and oracle have the same number of noise draws and sibling-sharing structure; they differ in whether noise is coupled to the factual outcome. Factual arrays are asserted exactly equal across arms. CartPole reward is fixed at one per retained transition, and the simulator recomputes the terminal label for every generated next state. Every learned arm uses the same initialization and number of updates. Generated arms sample fixed 50:50 real/generated minibatches of size 256, so augmentation does not increase optimizer steps or minibatch size.

The local follow-up manifest fixes seeds 1030–1059 before execution. Its primary contrast is oracle minus fresh-shared clean return. Secondary contrasts are the same comparison under process noise and fresh-shared minus fresh-independent under both evaluations. The study-defined operational thresholds are 25 clean-return points and 5 process-noise points; they are design choices, not externally validated standards.

3.4 Learned-SCM gate

We also implement a positive-weight BiCoGAN candidate with the layer widths reported by Lu et al. The generator is coordinatewise nondecreasing in its four-dimensional latent; this is not claimed to prove strict identifiability. It is trained on all factual trials and evaluated on an independent 50-trial bank. Before downstream use, all five development seeds must satisfy: learned-CF normalized mean-squared error (MSE) at least 20% below fresh-noise MSE, terminal disagreement below 5%, every latent standard deviation in $[0.5, 2]$, and action reconstruction MSE at least 10% below a central-action predictor.

3.5 Statistical analysis

We report seed-level mean and standard deviation, median, paired deltas, 95% paired bootstrap intervals, paired t , Wilcoxon, and sign-randomization tests, and Holm-adjusted values across the four manifest-specified contrasts. The raw

primary test is reported because that contrast was fixed first; adjusted values are a familywise sensitivity analysis. A practical-equivalence label requires the paired 90% bootstrap interval to lie within its study-defined band. Given bounded, ceiling-heavy RL returns, paired distributions and sensitivity across tests are emphasized over a single p -value [1].

3.6 AI-assisted workflow disclosure

We used OpenAI Codex with GPT-5.6 Luna (medium reasoning) and GPT-5.6 Sol (high reasoning), as identified by the run metadata, accessed August 31–September 1, 2026, for experiment-audit planning, code and test scaffolding, statistical cross-checks, citation verification, and language editing. The following is a condensed initiating directive; the supplementary disclosure preserves the verbatim prompt:

“Can you create create a plan on whatever improvements we can right now. Run them using subagents using luna model as its cheaper to run through them and use them whenever low reasoning is required. Then use sol at high effort to analysis and plan next move until you can’t think of anything and we reached at point where we can edit the latex with comments we got before it gets published.”

The supplementary disclosure records the focused audit prompts and their roles. OpenAI Deep Research and Google Deep Research critiques supplied by the authors were also used to identify literature, protocol, and reporting risks. Their verbatim initiating prompts and underlying model identifiers were not present in the supplied records; the supplementary disclosure records this limitation and gives non-verbatim scope reconstructions. No reported number was taken from either critique: retained claims, citations, and result artifacts were checked against primary sources or included raw files. No generative image model was used; all figures are deterministic plots of the included result artifacts. The authors inspected the source, raw artifacts, statistical recomputations, references, and rendered manuscript and retain full responsibility for the design, claims, citations, and submitted text.

4 Results

4.1 The reported-architecture reproduction fails

Lu et al. provide CartPole-SD results only in Figure 2(a). Vector-path inspection, rounded to the nearest ten, gives approximately 260 for D3QN and 310 for CTRLg at 5,000 samples. Their models train to convergence and are tested for ten trials, but training-seed uncertainty and evaluation noise are not given. Table 2 therefore gives descriptive context only.

Table 2. Protocol-qualified comparison with Lu et al. Plot values are approximate; NR means not reported, and no cross-paper statistical comparison is made.

Setting	Train seeds	Reported return	Protocol/budget note
Lu D3QN	NR	≈ 260	5,000 samples; eval. noise NR
Lu CTRLg	NR	≈ 310	BiCoGAN CF; eval. noise NR
Our reported-architecture D3QN	5	31.19 ± 6.84	mean 3,556 transitions; clean eval.
Our stabilized real-only ($2 \times 256 + \text{CQL}$)	30	266.09 ± 162.59	5,000 updates; clean eval.

A ten-seed protocol matrix does not repair the reported network: clean means range from 18.35 to 34.56 across the state-noise and trial-continuation variants. Continuing every trial to exactly 20 transitions changes the process-noise clean mean from 32.34 to 34.56. In contrast, a ten-seed online DQN control, selected on a separate validation seed bank and reported on a disjoint test bank, reaches 431.00 ± 66.12 clean and 46.88 ± 3.21 under process noise, versus random 27.15 ± 0.96 and 17.86 ± 0.52 . The environment is learnable; the failure is specific to the attempted offline reproduction.

4.2 Initial oracle study motivates a matched control

In an initial paired 30-seed simulator study, oracle CF scored 55.40 points below fresh-independent synthetic augmentation on clean evaluation (95% paired bootstrap interval $[-98.94, -9.54]$). That comparison changed both factual-noise coupling and the number and dependence of noise draws among ten synthetic siblings. We therefore treat it as motivating evidence, not the primary mechanism result, and designed the sibling-sharing-matched follow-up below. The two cohorts are not pooled.

4.3 Matched oracle coupling study

Table 3 reports the follow-up’s 30 distinct training/data seeds. The stabilized real-only arm is in the numerical range of Lu et al.’s plot, but its large dispersion and changed learner preclude a reproduction claim.

Table 3. Follow-up return across 30 seeds (mean $\pm$ SD; median).

Arm	Clean	Process noise
Random	27.03 ± 1.52 ; 27.20	17.44 ± 0.99 ; 17.60
Real only	266.09 ± 162.59 ; 224.48	27.51 ± 4.81 ; 27.53
Fresh-independent	322.27 ± 139.54 ; 292.78	37.94 ± 2.85 ; 37.98
Fresh-shared	367.15 ± 140.23 ; 429.66	37.34 ± 3.49 ; 38.25
Oracle CF	304.37 ± 136.75 ; 279.50	35.29 ± 3.52 ; 35.58

Table 4 reports every manifest-specified contrast rather than only the primary comparison. The three p -values in each cell are from the paired t , Wilcoxon, and sign-randomization tests, respectively.

Table 4. Paired follow-up contrasts over 30 training seeds. Intervals are 95% paired bootstrap intervals; p columns list t /Wilcoxon/randomization.

Contrast	Mean Δ	95% interval	Raw p	Holm p
Oracle — fresh-shared, clean	−62.78	[−111.31, −14.56]	.0193/.0080/.0192	.0580/.0241/.0575
Oracle — fresh-shared, process	−2.05	[−2.93, −1.13]	< .001/ < .001/ < .001	< .001/.003/.001
Fresh-shared — independent, clean	44.88	[−16.46, 103.68]	.161/.0775/.161	.321/.155/.322
Fresh-shared — independent, process	−0.60	[−1.49, 0.31]	.213/.221/.210	.321/.221/.322

Figure 1 shows that the arm means conceal heterogeneous, ceiling-heavy seed outcomes; paired comparisons are therefore more informative than unpaired differences between these marginal summaries.

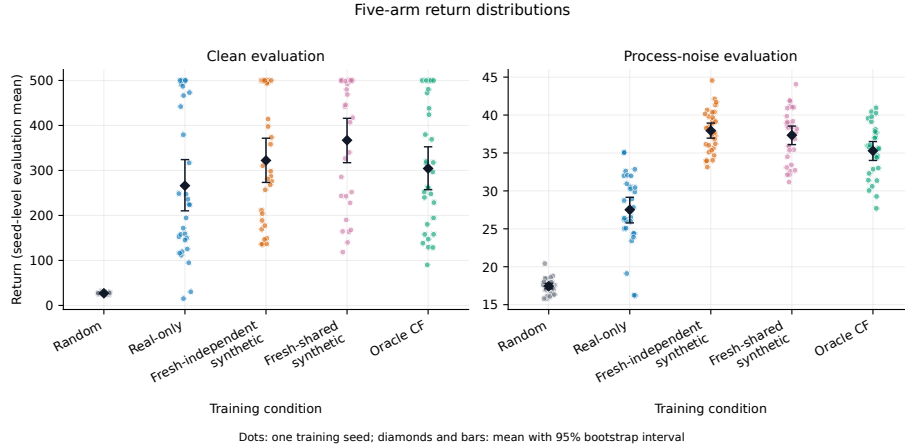


Fig. 1. Seed-level returns for the matched follow-up. Points are training-seed means; diamonds and bars show mean and 95% bootstrap interval. The evaluation scales are separated.

The primary oracle-minus-fresh-shared clean difference is -62.78 (95% paired bootstrap interval $[-111.31, -14.56]$). Oracle is lower in 20 seeds, higher in 5, and tied in 5; the paired median is -47.63 . Raw t , Wilcoxon, and randomization p -values are .0193, .0080, and .0192. After Holm adjustment, they are .0580,

.0241, and .0575. Thus, the direction is consistent, but familywise significance is test-dependent, and the interval does not establish that harm exceeds the study-defined 25-point operational threshold.

Under process noise, oracle is 2.05 points lower than fresh-shared (95% bootstrap interval $[-2.93, -1.13]$) and its 90% interval lies within $[-5, 5]$. It therefore meets our study-defined practical-equivalence rule while remaining statistically lower. Fresh-shared minus fresh-independent is inconclusive on clean evaluation (44.88, 95% interval $[-16.46, 103.68]$) and also meets the ± 5 rule under process noise (-0.60 , 95% interval $[-1.49, 0.31]$).

Figure 2 displays every seed-level primary difference and makes the sign imbalance, ties, and wide effect distribution visible.

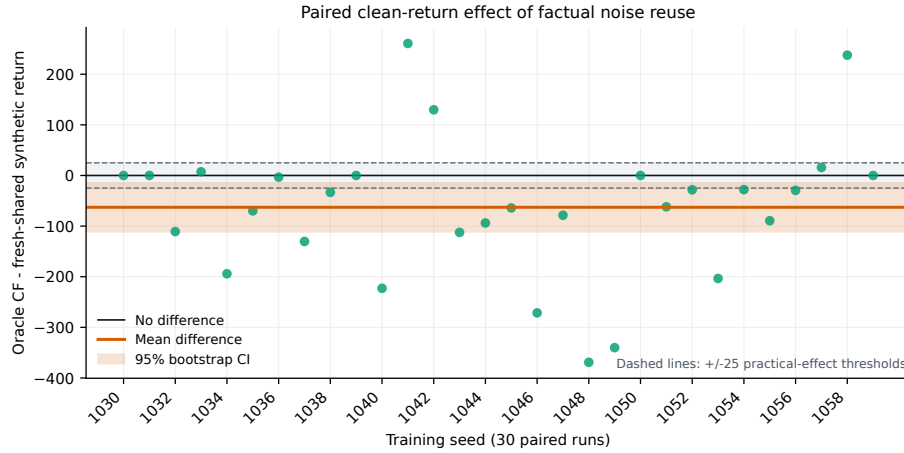


Fig. 2. Paired oracle-minus-fresh-shared clean-return differences. The solid line and band show the mean and 95% paired bootstrap interval; dashed lines mark the study-defined ± 25 practical thresholds.

4.4 The learned SCM fails its gate

The learned-CF normalized MSE is 21.4–29.5 times the fresh-noise-to-oracle MSE across five development seeds. Terminal disagreement is 42.0–65.9%, every seed has at least one latent dimension outside its accepted range, and action reconstruction passes in only three seeds. The supplementary archive includes the per-seed transition-fidelity and termination-consistency diagnostic. No learned-CF policy is therefore reported.

Table 5. Held-out BiCoGAN development gates. Eligibility required every gate in every seed; normalized MSE is relative to state scale.

Metric	Gate	Observed seeds	across Passed
Learned/fresh MSE	normalized < 0.8	21.4–29.5	0/5
Terminal disagreement	< 0.05	0.420–0.659	0/5
Latent SD, each dimension	$[0.5, 2.0]$	at least one viola- tion/seed	0/5
Action/baseline MSE	< 0.9	0.863–0.931	3/5

Table 5 makes the aggregate decision explicit: three gates fail in every seed, while action reconstruction alone passes in three seeds.

5 Discussion and Limitations

The matched follow-up rules out one explanation for the initial result: fresh augmentation did not win solely because it used ten independent sibling noise draws. Even after matching sharing, factual-noise reuse has lower clean mean return. This supports a narrow conclusion: exact counterfactual coupling is not sufficient to outperform fresh-shared augmentation for this learner, dataset, and mixing rule. It does not imply that counterfactual augmentation is generally harmful. Both synthetic controls use the true simulator, whereas practical CTRL must estimate the SCM, reward, and termination behavior.

The matched contrast is still an operational comparison, not a pure test of causal identifiability. Reusing factual noise also couples each synthetic sibling to the real transition present in training, whereas fresh-shared noise does not. The effect can therefore include real–synthetic redundancy and changes in effective sample size. Fresh-shared and fresh-independent also begin from the same random-number stream but consume it differently, leaving one identical synthetic next state per seed (about 0.003% of either pool). This negligible overlap cannot explain the reported contrasts, but distinct stream offsets would be preferable in a future rerun. In the follow-up cohort, the unregistered oracle-minus- fresh-independent clean difference was only -17.90 and was inconclusive; we do not describe the follow-up as a replication of the initial effect.

Several limitations remain. First, the original paper’s failure, convergence, mixing, and evaluation details are unavailable, preventing exact attribution of the reproduction gap. Second, CQL and the compact network make the mechanism study intentionally different from CTRL. Third, clean returns are ceiling-heavy and variable, motivating paired results and sensitivity tests. Fourth, our positive-weight generator is nondecreasing rather than proven strictly monotone, and its empirical failure cannot adjudicate the CTRL theorem or BiCoGAN generally. Finally, this revision removes prior LunarLander, MuJoCo, SAC, and

D4RL rows: they were underpowered, lacked a valid continuous-action CF definition, or did not contain a CF condition. Broad external-validity claims require new, compute-matched studies after the learned SCM passes task-specific state, reward, and termination gates.

6 Conclusion

Our audit does not reproduce learned CTRL. It shows why environment learnability, exact counterfactual coupling, and learned-model validity must be tested separately. A stabilized learner reaches nontrivial CartPole performance, but exact factual-noise reuse is lower than a matched fresh-shared simulator control on clean evaluation and meets the study-defined process-noise practical-equivalence criterion. The attempted learned SCM fails independent validation by a wide margin. These negative results replace the earlier broad benefit claim with a reproducible boundary: under the tested protocol, noise reuse provides no advantage over the matched fresh-shared simulator control, and learned augmentation should not proceed without explicit counterfactual-quality gates.

Artifact availability. The artifact is available in the versioned public project archive. It contains frozen manifests, source hashes, exact commands, seed-level JSON artifacts, summarizers, tests, and figure scripts. Because the remote checkout did not expose Git metadata, the archive itself is the authoritative source snapshot.

A Protocol Sensitivity Details

Table 6 reports the full matrix behind the protocol range in Section 4.1. The matched-noise evaluation uses the state-noise semantics named in each row; it is not one common process-noise evaluation.

Table 6. Development-only protocol matrix over ten seeds (mean $\pm$ SD).

State noise	Trial rule	Transitions	Clean	Matched noise
Process	Stop	3,494	32.34 ± 18.16	18.83 ± 2.41
Process	Continue	5,000	34.56 ± 11.07	19.13 ± 1.84
Observation	Stop	4,663	18.35 ± 5.72	17.97 ± 5.33
Observation	Continue	5,000	21.42 ± 4.08	20.57 ± 3.34

“Continue” advances the custom dynamics after the pre-noise failure boundary to preserve 20 consecutive transitions; it is a sensitivity analysis, not a claim about the unpublished original implementation. Process versus observation noise under continuation changes clean return by 13.14 points (95% paired t interval $[6.62, 19.66]$), but neither interpretation approaches the original plot with the reported-architecture learner.

B Initial Oracle Study

Table 7 gives the marginal arm summaries from the first 30-seed exact-simulator study. Its oracle-versus-fresh-independent comparison motivated the matched follow-up, but the design changed both factual coupling and sibling-noise diversity. The cohorts are therefore reported separately and are not pooled.

Table 7. Initial exact-simulator study over 30 training seeds (mean $\pm$ SD).

Arm	Clean Process noise	
Random	26.93 ± 1.58	17.08 ± 1.02
Real only	272.90 ± 165.71	28.21 ± 5.13
Fresh-independent	335.49 ± 122.94	38.61 ± 2.94
Oracle CF	280.08 ± 119.55	36.73 ± 3.92

Oracle minus fresh-independent clean return was -55.40 (95% paired bootstrap interval $[-98.94, -9.54]$), with 21 negative, 7 positive, and 2 tied seed differences. In the later follow-up cohort, the analogous unregistered oracle-minus-fresh-independent difference was only -17.90 and inconclusive. This heterogeneity reinforces why the fresh-shared comparison, rather than a pooled post-hoc estimate, is the primary mechanism result.

C Artifact Integrity and Model Diagnostics

The follow-up contains every seed from 1030 through 1059 exactly once. Every artifact records the same manifest hash (prefix `0dbb740b`), source digest (prefix `b9b2c137`), combined source/manifest digest (prefix `d140ab2b`), configuration, and software environment. Factual state, action, reward, next-state, terminal, and trial-index arrays are asserted identical before constructing the three synthetic pools. For fresh-shared and oracle CF, the maximum within-transition action- and state-noise reuse error is zero in every run. Each synthetic pool contains exactly ten transitions per factual transition.

The archive includes all raw JSON files and validators. From a clean environment, 50 focused tests pass; the four principal summaries regenerate byte-for-byte; and both PDF and PNG figures regenerate from the stored summaries. Git metadata was unavailable on the remote execution checkout, so the exact archived files and their hashes, rather than a remote commit field, bind code to results.

Figure 3 shows the two largest learned-model failures. The transition ratio is more than 21 rather than below 0.8, and terminal disagreement is at least 0.42 rather than below 0.05. These are development diagnostics, not downstream confirmatory evidence.

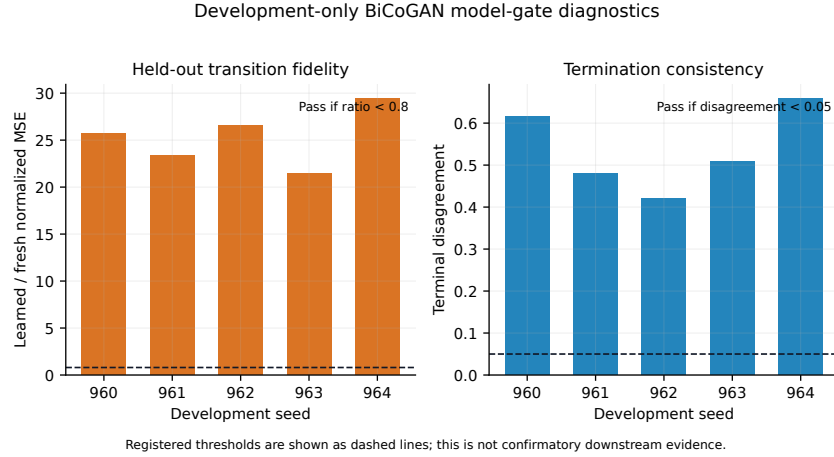


Fig. 3. Held-out development diagnostics for the attempted BiCoGAN. Dashed lines are manifest-specified gates, not fitted decision thresholds. Both shown metrics fail every seed.

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